

AS91946: Interpret and Apply Mathematical and Statistical Information

NCEA Level 1 Statistics — Year 11

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as to finish the design etc
before writing a probability.

Keep this for
now.

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1 Welcome

① Remove R code from pdf only.

② Render a single R script with all analysis and plotting etc.

Welcome

This resource has been developed to support Year 11 students preparing for NCEA Level 1 Achievement Standard 91946: *Interpret and apply mathematical and statistical information in context* (4 credits).

1.1 What this standard is about

In this standard you will learn to:

- Read and interpret statistical displays, tables, and graphs
- Apply mathematical and statistical ideas to real-world problems
- Communicate findings clearly using statistical language
- Evaluate statistical claims made in media and everyday life

Two standards

1.2 How to use this book

Each chapter builds on the previous one. You will find:

- Clear explanations of key concepts with New Zealand examples
- Interactive R code blocks showing how statistics is done with real data
- Practice questions at Achievement, Merit, and Excellence levels
- Glossary boxes highlighting important vocabulary

where to find official info... ncea

Achievement Standard at a Glance

Grade	What you need to do
Achieved	Interpret and apply statistical information
Merit	Interpret and apply statistical information with justification
Excellence	Interpret and apply statistical information with statistical insight

1.3 Tools we use

All data visualisations and calculations in this book are produced using R and rendered with Quarto. You do not need to be able to write R code to succeed in this standard — the code blocks are provided so your teacher can demonstrate concepts live and so you can explore data yourself if you choose.

Discussion in final chapter...

for correcting

Average Monthly Temperature — Wellington, NZ (°C)

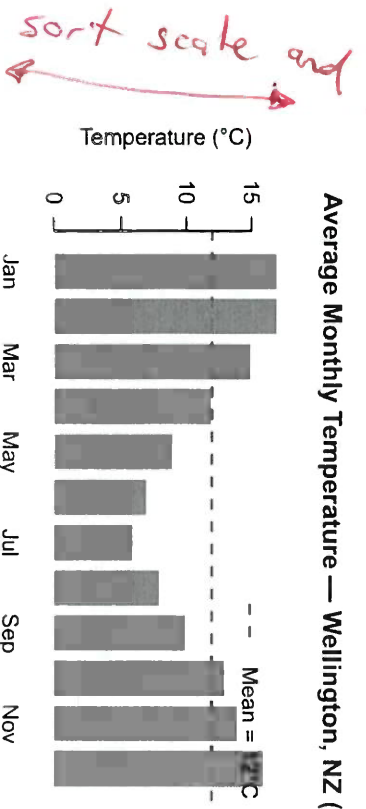


Figure 1.1: An example of the kind of visualisations we will create in this book.

1.4 Getting started

Head to Chapter 3 to begin your statistics journey!

Independent examples used through the book.

create diagram of workflow including R code and plot here just one

||| sidebar => there are four templates created by 'xx' to work from if you like.

Key resources if this isn't your thing

- Khan academy
- Statista...

2 What is Statistics?

Cartoon of lies, lies, statistics

image...

IMACE HEADER

3 What is Statistics?

3.1 Why statistics matters

Every day we are surrounded by data. News headlines cite poll results, sports commentators compare player averages, health campaigns report risk percentages, and social media shows us trending topics. Statistics is the science that helps us make sense of all this information.

! Key definition

Statistics is the practice of collecting, organising, summarising, analysing, and interpreting data in order to draw conclusions and make decisions.

3.2 The statistical enquiry cycle (PPDAC)

New Zealand's statistics curriculum uses the PPDAC cycle as a framework for carrying out a statistical investigation:

Stage	What you do
Problem	Pose a statistical question
Plan	Plan how to collect data
Data	Collect and record data
Analysis	Explore and summarise the data
Conclusion	Interpret results and answer the question

Colouring poster ooo

3 What is Statistics?

The PPDAC Cycle

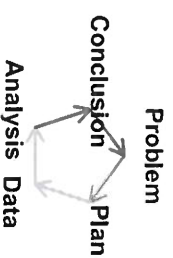


Figure 3.1: The PPDAC cycle — a framework for statistical investigations

3.3 Types of statistical questions

Not all questions are statistical questions. A statistical question expects variability in the answers.

! Tip #1

Statistical question: "How tall are the students in Year 11?"
(Heights will vary — this is statistical!)

Not a statistical question: "How tall is Ms Smith?"
(Only one answer — not statistical.)

! Tip #2

split into two boxes
open during

3.3.1 Practice identifying statistical questions

Which of the following are statistical questions?

1. How many pages are in Harry Potter and the Philosopher's Stone?
2. How many hours per week do Year 11 students spend on homework?
3. What is the population of New Zealand?
4. What are the most popular music genres among teenagers in Auckland?
5. What colour is the school uniform?

! Answers

1. Not statistical — there is one fixed answer.
2. Statistical — different students spend different amounts of time.
3. Not statistical — there is one (changing but definite) answer.
4. Statistical — different teenagers have different preferences.

+ 5 more spread out to fit on one page

- Not statistical — the uniform colour is fixed.

3.4 Why context matters

In AS91946, context is central. You will always work with data that relates to real situations. When interpreting statistics, you must:

- Name the variable being measured
- State the units (e.g. centimetres, dollars, minutes)
- Refer to the **population** the data came from
- Relate findings back to the real-world situation

▲ Common mistake

Saying "the mean is 24" is not enough. Say
"The mean age of survey respondents was 24 years."

3.5 Glossary

Term	Meaning
Data	Values collected by observation or measurement
Variable	A characteristic that can take different values
Population	The entire group of interest
Sample	A subset of the population selected for study
Statistic	A numerical summary calculated from a sample
Parameter	A numerical summary of a whole population

3.6 Check your understanding

- In your own words, explain what the PPDAC cycle is.
- Give two examples of statistical questions you could investigate at your school.
- A student says "the average score is 65". What is missing from this statement?
- What is the difference between a population and a sample?

Q. 1
Qualitative vs. Quantitative

4 Types of Data and Variables

Extend this out 2x pages including practice Q's.
Population vs sample

Ext tickets: Create styled box and all answers hidden.

5 Types of Data and Variables

Understanding the **type of data** you have is the first step in choosing the right graph or summary statistic. The wrong choice can make data hard to understand — or even misleading.

5.1 Categorical vs numerical data

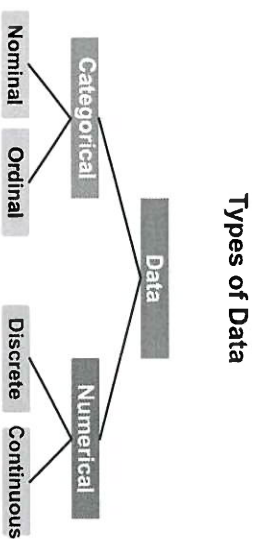


Figure 5.1: Classification of data types.

5.2 Categorical data

Categorical (qualitative) data describes qualities or categories. The values are labels, not numbers.

5.2.1 Nominal data

Nominal data has **no natural order**. Categories are just names.

Examples: - Eye colour (brown, blue, green, grey) - Favourite sport (rugby, netball, football, basketball) - Type of transport to school (bus, car, walk, bike)

5.2.2 Ordinal data

Ordinal data has a **natural order**, but the gaps between categories are not equal.

Examples: - Survey ratings (strongly disagree → disagree → neutral → agree → strongly agree) - Year group (Year 9, Year 10, Year 11, Year 12, Year 13) - Clothing size (XS, S, M, L, XL)

5.3 Numerical data

Numerical (quantitative) data represents quantities that can be measured or counted.

5.3.1 Discrete data

Discrete data can only take **specific, separate values** (usually whole numbers from counting).

Examples: - Number of siblings - Number of text messages sent per day - Goals scored in a rugby match

5.3.2 Continuous data

Continuous data can take **any value within a range** (usually from measuring).

Examples: - Height (cm) - Weight (kg) - Time to complete a race (seconds) - Temperature (°C)

5.4 Real data example

```

# Simulated class survey data
set.seed(7)
n <- 10

survey <- data.frame(
  Student = paste0("S", 1:n),
  Gender = sample(c("Male", "Female", "Non-binary"), n, replace = TRUE),
  Height_cm = round(rnorm(n, 168, 10), 1),
  Siblings = sample(0:4, n, replace = TRUE),
  Transport = sample(c("Bus", "Car", "Walk", "Bike"), n, replace = TRUE),
  Rating = sample(c("Poor", "Fair", "Good", "Excellent"), n, replace = TRUE)
)

knitr::kable(survey, align = "c")
  
```

Table 5.1: Example student survey data — identify the type of each variable.

Student	Gender	Height_cm	Siblings	Transport	Rating
S1	Female	154.3	1	Bus	Fair
S2	Non-binary	144.2	3	Bike	Excellent
S3	Non-binary	163.2	2	Bus	Excellent
S4	Non-binary	162.6	3	Bus	Poor
S5	Female	181.2	0	Car	Excellent
S6	Non-binary	152.9	2	Bike	Fair